

Engineering Document E10 — Billing & Subscription Architecture

Product Name: HQ

Engineering Package: Version 1.0

Purpose

This document defines the complete billing, subscription, entitlement, and monetization architecture for HQ.

Rather than limiting users based on subscription names, HQ uses an **Entitlement-Based Access Control (EBAC)** model. Every feature is protected by capabilities (entitlements), and subscription plans simply enable different capability sets.

This architecture supports the hackathon MVP while providing a scalable foundation for future commercial growth.

Design Principles

The billing system must be:

- Subscription-driven
- Entitlement-based
- Secure
- Auditable
- Scalable
- Extensible
- Provider-independent

Billing logic must remain separate from business logic.

Billing Architecture

User
↓
Organization
↓
Subscription
↓
Plan
↓
Entitlements
↓
Feature Access

Every feature request is validated against organization entitlements.

Core Billing Entities

Organization

↓

Subscription

↓

Plan

↓

Entitlement

↓

Invoice

↓

Payment

↓

UsageRecord



WebhookEvent

Subscription Plans (Initial)

Free

Designed for evaluation.

Includes:

- Limited monthly missions
 - Limited AI usage
 - Basic Executive Board
 - Standard templates
-

Pro

Designed for creators and professionals.

Includes:

- Higher AI limits
 - Full Executive Board
 - Content Studio
 - Asset Library
 - Advanced analytics
-

Team

Designed for agencies and growing businesses.

Includes:

- Multiple members
- Team collaboration

- Shared assets
 - Organization analytics
 - Higher quotas
-

Enterprise

Designed for large organizations.

Includes:

- Unlimited members
 - Priority AI processing
 - Custom integrations
 - Dedicated support
 - Enterprise security
 - Custom entitlements
-

Entitlement Model

Features are protected by capability keys.

Examples:

`missions.create`

`missions.delete`

`missions.export`

`executives.ceo`

`executives.finance`

`executives.legal`

`executives.security`

`assets.upload`

`assets.export`

`analytics.basic`

`analytics.advanced`

billing.manage

organization.invite

team.members

content.generate

content.publish

ai.image.generate

ai.video.generate

Plans enable collections of entitlements.

Feature Access Flow

Request

↓

Authentication

↓

Organization

↓

Subscription

↓

Entitlement Check

↓

Allow / Deny

Authorization is capability-driven.

Usage Limits

Some entitlements include quotas.

Examples:

- Monthly AI generations
- Storage (GB)
- Mission executions
- Team members
- API requests
- Asset uploads

Usage resets according to the billing cycle.

Subscription Lifecycle

States:

Trial

↓

Active

↓

Past Due

↓

Suspended

↓

Cancelled

↓

Expired

State transitions are recorded in the audit log.

Trial Period

New organizations receive a configurable trial.

Default:

- 14 days
- Access to selected premium features

At trial expiration:

- Downgrade to Free
 - Preserve user data
 - Restrict premium capabilities
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Payment Provider

Version 1:

- Stripe

Future support:

- Google Play Billing
- Apple In-App Purchase
- Paystack
- Flutterwave
- Paddle

Payment providers are abstracted behind a billing service.

Checkout Flow

Choose Plan

↓

Checkout

↓

Payment

↓

Webhook Verification

↓

Subscription Activated

↓

Entitlements Updated

↓

Confirmation

Webhooks

Supported events:

- Subscription created
- Subscription renewed
- Subscription cancelled
- Payment succeeded
- Payment failed
- Refund processed

Webhook signatures must always be verified.

Invoice Management

Organizations can:

- View invoices
- Download invoices
- View payment history
- Update billing information

Invoices remain immutable after issuance.

Organization Billing

Only users with appropriate permissions can:

- Upgrade plans
- Cancel subscriptions
- Manage payment methods
- View invoices
- Purchase add-ons

Permissions are enforced through RBAC.

Add-On System

Organizations can purchase optional capabilities.

Examples:

- Additional AI credits
- Extra storage
- Additional team seats
- Premium executive packs
- Advanced analytics
- Future marketplace plugins

Add-ons integrate with the entitlement engine.

Usage Tracking

Track:

- AI requests
- Mission executions
- Storage usage
- API usage

- Team seats
- Asset generation

Usage data is available through the Analytics Center.

Notifications

Notify organizations when:

- Trial is ending
- Payment succeeds
- Payment fails
- Subscription renews
- Quota reaches 80%
- Quota reaches 100%

Notifications appear in-app and by email.

Security

Billing security includes:

- Signed webhooks
- Secure payment tokens
- Audit logging
- Permission checks
- Fraud detection hooks (future)

HQ never stores raw payment card information.

Database Relationships

Organization

1 → 1 Subscription

Subscription

1 → 1 Plan

Plan

1 → Many Entitlements

Organization

1 → Many Invoices

Organization

1 → Many Usage Records

API Endpoints

Billing module exposes:

- List plans
- Current subscription
- Upgrade plan
- Downgrade plan
- Cancel subscription
- Resume subscription
- Usage summary
- Invoice history
- Payment methods

All endpoints require authenticated access.

Analytics

Billing metrics include:

- Monthly Recurring Revenue (MRR)
- Annual Recurring Revenue (ARR)

- Trial conversion rate
- Churn rate
- Average Revenue Per User (ARPU)
- Feature adoption
- Entitlement utilization

These metrics support product and business decisions.

Future Expansion

The architecture supports:

- Regional pricing
- Promotional codes
- Referral rewards
- Enterprise contracts
- Seat-based licensing
- Usage-based billing
- Marketplace purchases

No major redesign should be required.

Success Criteria

The Billing & Subscription Architecture succeeds when:

- Feature access is controlled through entitlements.
 - Plans can evolve without code changes.
 - Billing remains secure and auditable.
 - Organizations understand their usage and limits.
 - The platform can grow from MVP to enterprise scale.
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Conclusion

The HQ Billing & Subscription Architecture establishes a flexible, entitlement-driven monetization system.

By separating subscription plans from feature capabilities, HQ gains the ability to evolve pricing, introduce new products, support enterprise customers, and launch future add-ons without restructuring the core application. This architecture provides a strong commercial foundation for long-term growth while remaining simple enough for the initial hackathon release.